

Syllabus

This course will teach a broad set of principles and tools that will provide the mathematical, algorithmic and philosophical framework for tackling problems using Artificial Intelligence (AI) and Machine Learning (ML). AI and ML are highly interdisciplinary fields with impact in different applications, such as, biology, robotics, language, economics, and computer science. AI is the science and engineering of making intelligent machines, especially intelligent computer programs, while ML refers to the changes in systems that perform tasks associated with AI. Ethical issues in advanced AI and how to prevent learning algorithms from acquiring morally undesirable biases will be covered.

Topics may include a subset of the following: problem solving by search, heuristic (informed) search, constraint satisfaction, games, knowledge-based agents, supervised learning (e.g., regression and support vector machine), unsupervised learning (e.g., clustering), dimension reduction, learning theory, reinforcement learning, transfer learning, and adaptive control and ethical challenges of AI and ML.

Course Learning Outcomes

- CLO1 Understand the fundamental concepts of computational intelligence and machine learning.
- CLO2 Demonstrate awareness of the major challenges and risks facing computational intelligence and the complexity of typical problems within the field.
- CLO3 Able to implement solutions to various problems in computational intelligence.

Topics covered

Course Content	No. of Hours	Course Learning Outcomes
Introduction	1	CLO1, CLO2
Uninformed Search, Informed Search, Local Search, Constraint Satisfaction Problems, Adversarial Search	8	CLO1, CLO2, CLO3
Markov Decision Processes, Reinforcement Learning	7	CLO1, CLO2, CLO3
Supervised and Unsupervised Machine Learning	7	CLO1, CLO2, CLO3
Philosophy of AI	1	CLO1, CLO2
Group Presentations	6	CLO3

Last modified: Thursday, 27 August 2026, 8:19 PM

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